WORKED SOLUTION

STEP III 1990, Question 9

Pure • Hyperbolic Functions

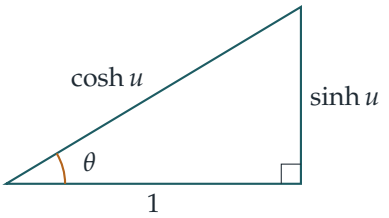
STEP III 1990, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q9

The key idea

The relation $\tan\theta = \sinh u$ is the same as $\sec^2\theta = 1 + \sinh^2 u = \cosh^2 u$, so the trigonometric and hyperbolic functions of the two variables match up exactly: $\cos\theta = \operatorname{sech} u$, $\sin\theta = \tanh u$, $\sec\theta = \cosh u$. A right-angled triangle with sides 1, $\sinh u$ and $\cosh u$ shows all of these at a glance. Once we have that dictionary, each part is a short differentiation followed by a translation back.



Solution

Throughout, since $0 \le \theta < \frac{1}{2}\pi$ we have $\tan\theta \ge 0$, so $\sinh u \ge 0$ and $u \ge 0$. Also $\cos\theta > 0$ and $\cosh u > 0$.

Part (i)

Using $1 + \tan^2\theta = \sec^2\theta$ and $1 + \sinh^2 u = \cosh^2 u$,

$$\sec^2\theta = 1 + \tan^2\theta = 1 + \sinh^2 u = \cosh^2 u.$$

Both $\sec\theta$ and $\cosh u$ are positive, so $\sec\theta = \cosh u$, that is, $\cos\theta = \operatorname{sech} u = v$, as required.

It is worth recording the companion fact: $\sin\theta = \tan\theta\cos\theta = \sinh u \operatorname{sech} u = \tanh u$.

Part (ii)

Differentiating $\tan\theta = \sinh u$ with respect to u (implicitly, by the chain rule),

$$\sec^2\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}u} = \cosh u \quad\Longrightarrow\quad \frac{\mathrm{d}\theta}{\mathrm{d}u} = \cosh u\cos^2\theta = \cosh u\,\operatorname{sech}^2 u = \operatorname{sech} u = v,$$

using part (i) for $\cos\theta$.

Part (iii)

Since $\frac{\mathrm{d}}{\mathrm{d}u}\cosh u = \sinh u$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{\mathrm{d}}{\mathrm{d}u}(\cosh u)^{-1} = -\frac{\sinh u}{\cosh^2 u} = -\operatorname{sech} u\tanh u.$$

With the dictionary, $\operatorname{sech} u\tanh u = \cos\theta\sin\theta = \frac{1}{2}\sin 2\theta$, so

$$\sin 2\theta = -2\,\frac{\mathrm{d}v}{\mathrm{d}u}.$$

For the second derivative, use the product rule together with $\frac{\mathrm{d}}{\mathrm{d}u}\tanh u = \operatorname{sech}^2 u$:

$$\begin{aligned}\frac{\mathrm{d}^2v}{\mathrm{d}u^2} &= -\Big[(-\operatorname{sech} u\tanh u)\tanh u + \operatorname{sech} u\cdot\operatorname{sech}^2 u\Big]\\ &= \operatorname{sech} u(\tanh^2 u - \operatorname{sech}^2 u).\end{aligned}$$

Multiplying by $-\cosh u$ cancels the $\operatorname{sech} u$:

$$-\cosh u\,\frac{\mathrm{d}^2v}{\mathrm{d}u^2} = \operatorname{sech}^2 u - \tanh^2 u = \cos^2\theta - \sin^2\theta = \cos 2\theta,$$

which is the second result.

Part (iv)

Now treat u and v as functions of θ. From part (i), $v = \cos\theta$ and $\cosh u = \sec\theta$, and part (ii) gives $\frac{\mathrm{d}u}{\mathrm{d}\theta} = 1/\frac{\mathrm{d}\theta}{\mathrm{d}u} = 1/v$ (valid because $v > 0$). So

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = \sec\theta, \qquad \frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} = \sec\theta\tan\theta,$$
$$\frac{\mathrm{d}v}{\mathrm{d}\theta} = -\sin\theta, \qquad \frac{\mathrm{d}^2v}{\mathrm{d}\theta^2} = -\cos\theta.$$

Ask yourself Before grinding through, does the identity in (iv) look exactly right as printed?

It does not. As printed, the first term has $\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}$, and then the left-hand side is

$$\begin{aligned}\sec\theta\cdot\sec\theta\tan\theta - \sin\theta\sec\theta\tan\theta + \sec^2\theta &= \sec^2\theta\tan\theta + (\sec^2\theta - \tan^2\theta)\\ &= \sec^2\theta\tan\theta + 1 \;\geq 1,\end{aligned}$$

which is never zero. The intended identity has $\frac{\mathrm{d}^2v}{\mathrm{d}\theta^2}$ in the first term (a misprint of u for v):

$$\frac{\mathrm{d}u}{\mathrm{d}\theta}\frac{\mathrm{d}^2v}{\mathrm{d}\theta^2} + \frac{\mathrm{d}v}{\mathrm{d}\theta}\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + \left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2 = 0.$$

Substituting the four derivatives,

$$\sec\theta(-\cos\theta) + (-\sin\theta)\sec\theta\tan\theta + \sec^2\theta = -1 - \tan^2\theta + \sec^2\theta = 0,$$

since $\sec^2\theta = 1 + \tan^2\theta$. This proves the corrected result.

A neater way to see it: the first two terms are $\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\frac{\mathrm{d}v}{\mathrm{d}\theta}\right)$ by the product rule, and $\frac{\mathrm{d}u}{\mathrm{d}\theta}\frac{\mathrm{d}v}{\mathrm{d}\theta} = -\sec\theta\sin\theta = -\tan\theta$, whose derivative is $-\sec^2\theta = -\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2$.

Checks

Take $u = \ln 2$, so $\sinh u = \frac{3}{4}$, $\cosh u = \frac{5}{4}$ and $\tan\theta = \frac{3}{4}$: the 3–4–5 triangle gives $\cos\theta = \frac{4}{5} = \operatorname{sech} u$ and $\sin 2\theta = \frac{24}{25}$, while $-2\,\mathrm{d}v/\mathrm{d}u = 2\cdot\frac{4}{5}\cdot\frac{3}{5} = \frac{24}{25}$. Numerically, finite-difference derivatives at 2000 random points agree with (i)–(iii) and the corrected (iv) to about 10^{-7}, and confirm that the printed left-hand side of (iv) equals $1 + \sec^2\theta\tan\theta$.

Common slips

- Taking $\cos\theta = \pm\operatorname{sech} u$ without saying why the positive sign is right: it is the range $0 \le \theta < \frac{1}{2}\pi$ that settles it.
- Forgetting that $\frac{\mathrm{d}u}{\mathrm{d}\theta}$ is just the reciprocal of $\frac{\mathrm{d}\theta}{\mathrm{d}u}$, and differentiating $\operatorname{arsinh}(\tan\theta)$ from scratch.

Beyond STEP

The function $\theta(u) = \arctan(\sinh u)$ is the *Gudermannian* function, which links circular and hyperbolic functions without complex numbers; its inverse $u = \ln(\sec\theta + \tan\theta) = \int_0^\theta \sec t\,\mathrm{d}t$ is exactly the vertical coordinate on a Mercator map. The same pairing appears when you write $\cosh(\mathrm{i}x) = \cos x$, and Needham's *Visual Complex Analysis* is a beautiful place to see these connections geometrically.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.